

A Calibrated, Explainable Detector for AI-Generated Images

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Abstract

We present a practical detector for AI-generated images built around three layered components: (i) a provenance check for embedded content-credential metadata (C2PA / SynthID), (ii) a forensic model that fuses a frozen SigLIP backbone with a custom hand-crafted artifact stream derived from the technical reports of modern generators, and (iii) a Neyman–Pearson calibrated decision rule with a controllable false-alarm rate and per-stream explanations. Evaluated honestly on the in-the-wild *Chameleon* benchmark—where many published detectors collapse to near chance—the fused model attains **0.90 AUC**; the custom forensic stream contributes +0.05 AUC and +13 points of recall at a 5% false-alarm rate over a strong SigLIP-only baseline. On two 2025–26 frontier generators never seen in training (GPT-Image, Nano Banana) the detector reaches 0.90–0.96 AUC zero-shot and recovers to 83–91% detection at a 1% false-alarm rate with only 25–50 labeled examples via few-shot adaptation, with the false-alarm rate held at target throughout. We are deliberate about scope: in-the-wild detection is unsolved, and robustness to unseen generators comes from few-shot adaptation, not zero-shot magic.

1 Introduction

The rapid improvement of text-to-image models has made synthetic media difficult to distinguish from real photographs, with consequences for misinformation, fraud, and trust in visual evidence. Detection is hard precisely because it must *generalize*: a detector trained on one set of generators routinely fails on new ones, and on “in-the-wild” images that have been curated and post-processed by humans. A recent sanity check [1] showed that nine off-the-shelf detectors all degrade sharply on such images, systematically misclassifying fakes as real.

This project builds a detector that is (a) *honest*—evaluated on a held-out in-the-wild benchmark with confidence intervals—(b) *calibrated*, with a controllable and recoverable false-alarm rate, and (c) *practical*, deployed as a public web demo and able to adapt to a brand-new generator from a handful of examples. We do not aim to beat industrial systems; we aim for a rigorous, defensible result that reflects the true state of the problem.

2 Background

Three threads inform the design. **Frozen-backbone detection:** UnivFD [2] showed that a linear probe on frozen CLIP features generalizes across generators; we use the stronger SigLIP backbone [3] in the same frozen spirit. **Calibration under shift:** recent work [4, 5] shows that a large part of cross-generator failure is a *mis-aligned decision threshold* rather than a representation failure, and that recalibration—even from unlabeled data—restores much of the lost reliability. **Hybrid**

forensic experts: the SOTA AIDE detector [1] fuses a semantic (CLIP) expert with a low-level frequency/noise expert; our forensic stream follows this philosophy but designs its features directly from the documented failure modes of specific generators (Nano Banana, GPT-Image, DALL·E) [6, 7, 8].

3 Method

The detector is a three-layer pipeline.

Layer 1 — Provenance. OpenAI (C2PA) and Google (SynthID + C2PA) embed content-credential metadata in their outputs. A lightweight scan flags this metadata when present; it is the most reliable signal for those generators but is defeated by screenshots or re-encoding, motivating the forensic fallback.

Layer 2 — Forensic model. We extract two complementary, *frozen* feature streams and fuse them:

- **Semantic:** the SigLIP vision encoder (frozen, pretrained on $\sim 10\text{B}$ image-text pairs), giving a 768-d embedding.
- **Forensic (66-d, hand-crafted):** three feature groups motivated by the generator papers—(1) *spectral* (radial FFT power, high-frequency energy ratio, spectral slope) targeting over-smoothing / loss of high-frequency detail and upsampling artifacts; (2) *colour* (saturation, vibrance, channel correlations) targeting non-physical colour shifts; (3) *noise residual* (a PRNU proxy) exploiting that real photographs carry camera sensor noise that synthetic images lack.

A small per-stream head feeds a learned linear combiner; only the heads and combiner are trained, on cached features, in minutes.

Layer 3 — Calibrated verdict. We set the decision threshold τ with a split-conformal Neyman-Pearson rule, calibrated on real images alone, giving a finite-sample guarantee $\Pr(\text{score}(X_{\text{real}}) > \tau) \leq \alpha$. Each verdict is accompanied by the signed per-stream contributions ($w_i \cdot \text{logit}_i$, which sum to the score), so the explanation is consistent with the decision.

4 Data

Training combines diversity along two axes—content type and generator family—so the model cannot shortcut on “faces” or “diffusion-only”: real images from ImageNet (GenImage), COCO and FFHQ; fakes from the eight GenImage generators plus StyleGAN2 faces. The set is balanced (20k real / 20k fake) and all images are re-encoded to a common JPEG quality so the model learns *generation* artifacts rather than source compression [9]. Evaluation uses the held-out *Chameleon* benchmark [1] (14,863 real / 11,170 fake), never seen in training, and two frontier generators (GPT-Image, Nano Banana, ~ 100 images each) for the generalization study.

5 Results

5.1 In-the-wild detection (Chameleon)

Table 1 reports held-out Chameleon performance. The fused model attains 0.90 AUC; adding the forensic stream to the SigLIP baseline improves both ranking (+0.048 AUC) and, importantly, recall at a deployable false-alarm rate (+0.126 at 5% FPR). These results are on format-normalized images, so they reflect genuine signal rather than a compression artifact.

Table 1: Held-out Chameleon (3 seeds). Forensic stream improves a strong baseline.

Model	AUC	TPR@1%FPR	TPR@5%FPR	Accuracy@ τ
SigLIP-only (linear probe)	0.851	0.143	0.375	0.713
SigLIP + forensic (fusion)	0.899	0.252	0.501	0.797

5.2 Generalization to unseen frontier generators

Table 2 and Figure 1 show zero-shot performance on two generators absent from training, and recovery under few-shot adaptation. Zero-shot AUC is already high (0.90 for GPT-Image, 0.96 for Nano Banana), empirically confirming the generators’ papers, which note these models are detectable via upsampling and over-smoothing artifacts. With 25–50 labeled examples, detection at a 1% false-alarm rate rises to 83–91%; the false-alarm rate stays pinned at the target $\alpha = 0.01$ at every budget.

Table 2: Frontier generators (unseen; 3 seeds; $\alpha = 0.01$). TPR is detection at 1% FPR.

	zero-shot AUC	TPR ($k=0$)	TPR ($k=10$)	TPR ($k=25$)	TPR ($k=50$)
GPT-Image	0.903	0.300	0.673	0.787	0.827
Nano Banana	0.963	0.573	0.793	0.900	0.907

5.3 Calibration behaviour

The conformal threshold calibrated on training-domain reals transfers reasonably to Chameleon (realized FPR 0.063 at target 0.05 for the probe). Under a strong *real-domain* shift (training on object/scene reals, testing on face reals) a controlled experiment shows the guarantee breaking (FPR: 0.01 $\rightarrow$ 0.27) and being restored to target ($\rightarrow$ 0.01) by recalibrating τ on unlabeled reals from the new domain alone—no fake examples required. This separates what unlabeled real data can fix (the false-alarm rate) from what it cannot (detection power, which requires labeled fakes, as the few-shot curve shows).

6 Discussion and limitations

In-the-wild AI-image detection is unsolved; ~ 0.90 AUC on Chameleon is competitive, not perfect, and recall at a strict false-alarm budget remains modest. Robustness to a new generator is obtained by few-shot adaptation on a small labeled set, which is the realistic operating model as generators proliferate—not a permanent zero-shot property. The frontier evaluation sets are small (~ 50 test

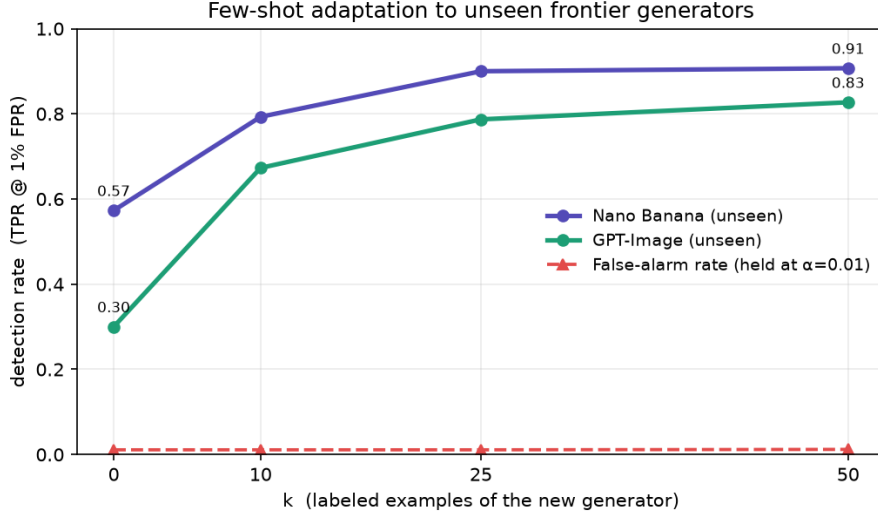


Figure 1: Few-shot adaptation to unseen frontier generators. Detection at a 1% false-alarm rate rises with a handful of labeled examples, while the false-alarm rate stays at the $\alpha = 0.01$ target.

images per generator), so those point estimates carry wide intervals although the trends are stable across seeds. The SigLIP backbone is Google’s, used frozen and credited, as is standard practice; the forensic stream, fusion head, calibration, provenance layer, data assembly and evaluation are contributions of this work.

7 Conclusion

We built and deployed a calibrated, explainable AI-image detector with an honest in-the-wild result (0.90 AUC on Chameleon), a measurable contribution from physics-motivated forensic features, and a demonstrated few-shot recipe for adapting to brand-new generators while holding the false-alarm rate at target. The system is available as a public, interactive demo.

References

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